Retail Analytics Capstone: From Transaction Data to Revenue Decisions

Usingretailtransactiondatatoimproveplanning,customerstrategy,and revenue performance

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Business Problem

Volatility

Retail demand fluctuates significantly across months and markets, creating planning uncertainty

Uneven Distribution

Revenue is concentrated among select customers and geographies, requiring strategic prioritisation

Decision Gap

Managers need data-backed insights for inventory management, promotional planning, and customer retention

Project Objective and Scope

End-to-End Workflow

Buildcomprehensiveanalyticspipeline from raw transaction data to dashboard-driveninsights using Python and Tableau

Key Analysis Areas

- Revenue trend analysis
- Top customer profiling
- Country performance mapping
- Seasonal patterns

Deliverable

Decision-readyoutputs supporting operational and strategic retail planning

KPI Snapshot

18.5K

Complete transaction records processed

5.2M

Units distributed across all markets

$8.9M

Generated across 37 countries

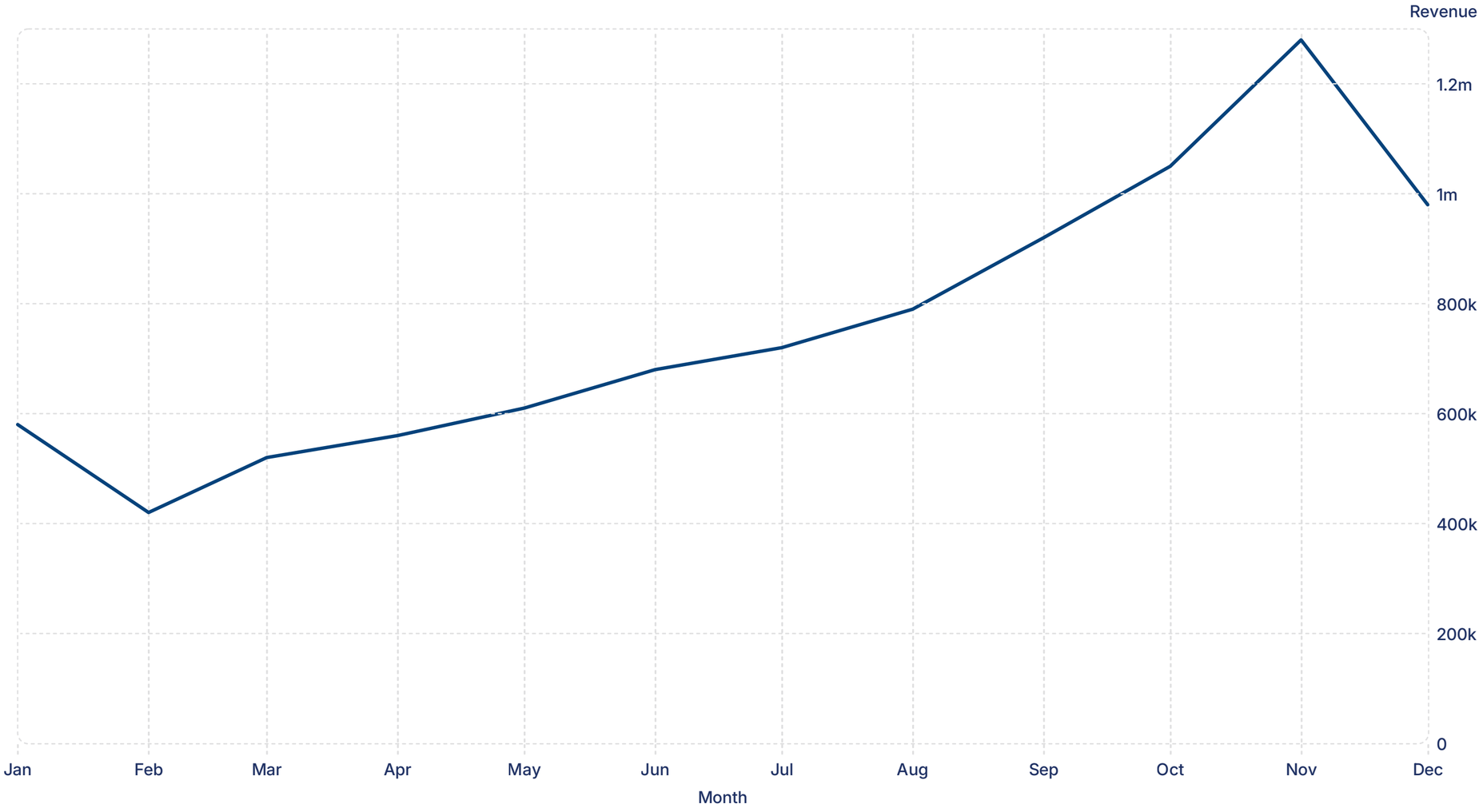
37

Global market footprint

Focus on where and when revenue originates for strategic planning

Revenue Trend Over Time

Monthly revenue patterns reveal strong seasonal demand cycles



Seasonality Story

Q4 Dominance

Fourth quarter drives primary revenue engine with holiday season demand surge

Peak Performance

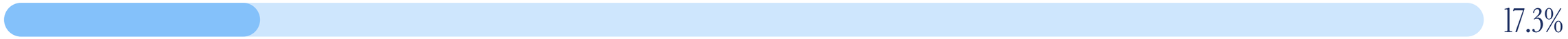
Novemberrevenuel59%higher than February low point

Planning Imperative

Pre-peakinventory planning essential; low-season campaigns needed for revenue smoothing

Customer Insights

Revenueconcentrationamongtopcustomerssuggests retention opportunities



Top 10 Customers

Contribution to total revenue from highest-value accounts

High-value customers require targeted retention programmes and personalised engagement strategies

Country Performance

TopMarkets

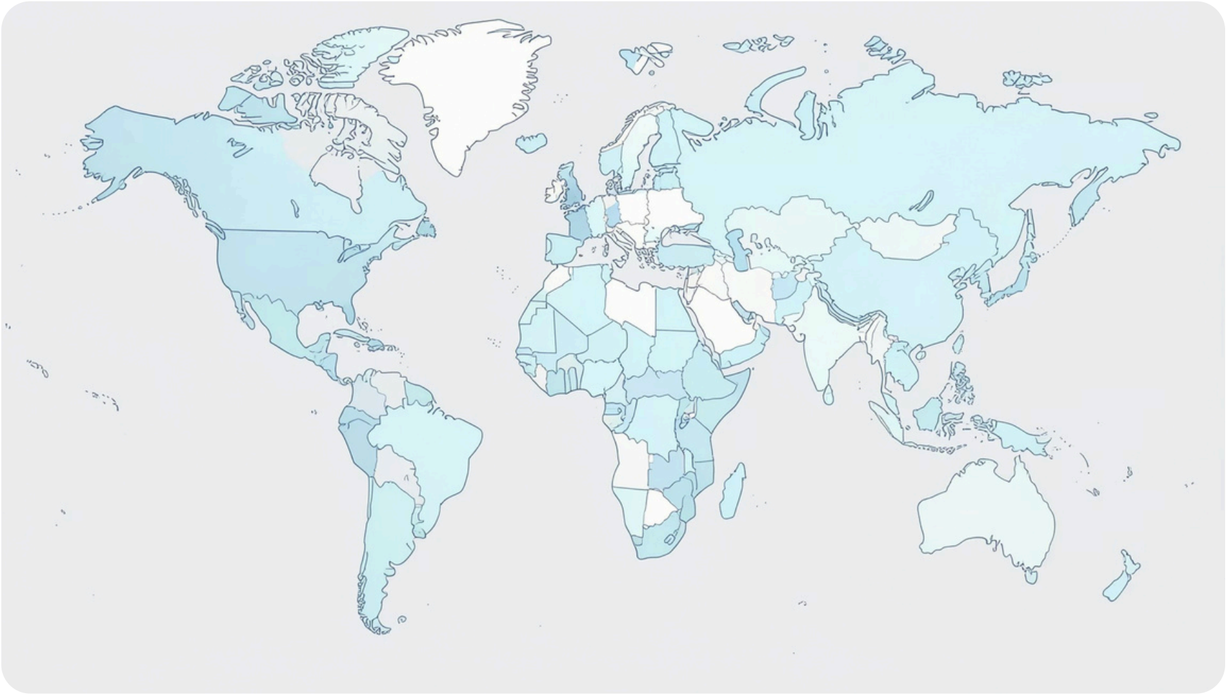
Netherlands,EIRE, and Germany lead revenue generation

Revenue Distribution

Significantvariationacross 37 countries creates opportunitiesforlocalized growth strategies

Strategic Implication

Geographicprioritizationand market-specific tactics recommended for revenue optimization



Strategic Recommendations

Seasonal Planning Calendar

Aligninventoryand staffingwith predicted demand patterns

Low-Season Campaigns

Develop promotionstosmooth revenue across quarters

Key-Customer Retention

Implementtargetedprogrammesfor top 10 customers

Top-Country Growth

FocusinvestmentonNetherlands, EIRE, and Germany

Dashboard Monitoring

DeployTableaudashboardsfor real-time tracking

Conclusion

From Reactive to Proactive

Analytics enablesforward-looking planning rather than backward-looking reporting

Improved Decision-Making

Data-driveninsightsenhance planning accuracy, customer strategy effectiveness, and market prioritization

Strategic Value

End-to-endworkflow delivers actionable intelligence for retail performance optimization

Thank you